

Maxwellovy rovnice ve 4D

Definice zdvojení

→ rozumná volba

$$J^\mu = (c\rho, \vec{j}) = (c\rho, \rho\vec{v}) = \frac{\rho}{\gamma} \gamma(c, \vec{v}) = \frac{\rho}{\gamma} u^\mu$$

- je to vektor? $\Leftrightarrow$ je $\frac{\rho}{\gamma}$ invariant?

$$\rho_0 \equiv \rho(v=0) = \frac{dQ}{dV_0} = \frac{dQ}{\gamma dV} = \frac{1}{\gamma} \rho$$

dQ je invariant? Zdá se, že ano

$$J^\mu = \rho_0 u^\mu$$

Definice potenciálu

→ nabízel se možnost $A^\mu = (\frac{\phi}{c}, \vec{A})$

- jestli je to čtyřvektor je potřeba ověřit, experimentálně přirozeně velice, protože potenciály nejsou měřitelné

Tenzor EM pole

→ přirozeně zavedeme $F_{\mu\nu} = A_{\nu,\mu} - A_{\mu,\nu}$

- $F_{\mu\mu} = 0$ triviálně
- $F_{0j} = A_{j,0} - A_{0,j} = \frac{\partial \vec{A}}{\partial ct} + \frac{1}{c} \nabla_j \phi = -\frac{1}{c} E_j$
- $F_{ij} = A_{j,i} - A_{i,j} = \epsilon_{ijk} \underbrace{E_{km}}_{(\vec{\nabla} \times \vec{A})^k} = \epsilon_{ijk} B^k \equiv B_{ij}$
 $-\epsilon_{ijk} E_{km} = -(\delta_{ik} \delta_{jm} - \delta_{im} \delta_{jk}) = -(A_{i,j} - A_{j,i})$

$$B_{ij} \equiv \epsilon_{ijk} B^k \Leftrightarrow B_i \equiv \frac{1}{2} \epsilon_{ijk} B^{jk}$$

→ veličina $F_{\mu\nu}$ obsahuje jak $\vec{E}$, tak $\vec{B}$

→ míchá je

→ je to tenzor? Je, pokud je A^μ tenzor.

Maxwellovy rovnice

$$\begin{aligned} \vec{\nabla} \cdot \vec{D} &= \rho & \vec{\nabla} \cdot \vec{B} &= 0 \\ \vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} &= 0 & \vec{\nabla} \times \vec{H} - \frac{\partial \vec{D}}{\partial t} &= \vec{j} \end{aligned} \quad \left. \vphantom{\begin{aligned} \vec{\nabla} \cdot \vec{D} &= \rho \\ \vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} &= 0 \end{aligned}} \right\} \text{klasicky}$$

1. Série $F^{\kappa\lambda}_{,\lambda} = \mu J^\kappa$ $\mu \dots$ permeabilita

$$F^{\alpha\lambda}_{,\lambda} = \underbrace{F^{00}}_{=0}_{,0} + F^{0i}_{,i} = \frac{1}{c} \vec{\nabla} \cdot \vec{E} = \mu c \rho \quad \left| \frac{1}{\mu c} = \epsilon \right.$$

$$\Rightarrow \vec{\nabla} \cdot \vec{D} = \rho$$

+ víme, že $F^{\mu\lambda}$ je vektor, protože jeho divergenci vzniká vektor

$$F^{\beta\lambda}_{,\lambda} = F^{\beta 0}_{,0} + F^{\beta i}_{,i} = -\frac{1}{c} \frac{\partial E_i}{\partial ct} + \epsilon^{\beta ik} \nabla_i B_k = \mu \vec{j}^\beta \quad \left| \frac{1}{\mu} = c\epsilon \right.$$

$$\Rightarrow -\frac{\partial \vec{D}}{\partial t} + \vec{\nabla} \times \vec{H} = \vec{j}$$

2. série $F_{\xi\mu\nu,\kappa\xi} = 0$

Plyno triviálně:

$$\cancel{A_{\nu,\mu\kappa}} - \cancel{A_{\mu,\kappa\nu}} + \cancel{A_{\mu\kappa,\nu}} - \cancel{A_{\kappa,\nu\mu}} + \cancel{A_{\kappa\nu,\mu}} - \cancel{A_{\nu,\mu\kappa}} = 0$$

Výsledek je totálně antisymetrický, protože

$$F_{\xi\mu\nu,\kappa\xi} = \underbrace{F_{\mu\nu,\kappa}}_{-F_{\nu\mu,\kappa}} + \underbrace{F_{\kappa\mu,\nu}}_{-F_{\mu\kappa,\nu}} + \underbrace{F_{\nu\kappa,\mu}}_{-F_{\kappa\nu,\mu}} = -F_{\xi\kappa\nu,\mu\xi}$$

$\Rightarrow$ netriviální jsou pouze

- $\xi\mu\nu\kappa = \{1,2,3\}$

$$F_{12,3} + F_{31,2} + F_{23,1} = \epsilon_{12k} B_{k,3} + \epsilon_{31k} B_{k,2} + \epsilon_{23k} B_{k,1} = \vec{\nabla} \cdot \vec{B} = 0$$

- $\xi\mu\nu\kappa = \{0,i,j\}$

$$F_{0i,j} + F_{j0,i} + F_{ij,0} = -\frac{1}{c} E_{i,j} + \frac{1}{c} E_{j,i} + \epsilon_{ij}^k \frac{\partial B_k}{\partial ct} \quad \left| \cdot c \epsilon_{ij}^k \right.$$

$$2\vec{\nabla} \times \vec{E} + 2\frac{\partial \vec{B}}{\partial t} = 0 \Rightarrow \vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0$$

Rovnice kontinuity

$$F^{\mu\nu}_{,\mu\nu} = 0$$

triviálně: $F^{\mu\nu}_{,\mu\nu} = -F^{\nu\mu}_{,\mu\nu} = -F^{\nu\mu}_{,\nu\mu} = -F^{\mu\nu}_{,\mu\nu}$

↓ Maxwelly

$$\text{Rovnice kontinuity } J^\mu_{,\mu} \Leftrightarrow \frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0$$

Kalibrační transformace

$$A^\mu \rightarrow \tilde{A}^\mu = A^\mu + \chi^\mu$$

$$\tilde{F}^{\mu\nu} = A^{\nu,\mu} - A^{\mu,\nu} + \underbrace{\chi^{\nu,\mu} - \chi^{\mu,\nu}}_{=0} \stackrel{!}{=} F^{\mu\nu}$$

$$\Rightarrow \chi^{\nu,\mu} \stackrel{!}{=} \chi^{\mu,\nu}$$

→ lze splnit, pokud

$$\chi^\mu = \underbrace{\chi_{,i}^\mu}_{\text{skalární funkce}}$$

- nejobvyklejší choice $A^\mu_{,\mu} = 0$

$$\Rightarrow \square \chi = \chi_{,i}^\mu{}_\mu = -A^\mu_{,\mu}$$